Radioactivity and particles

Mark scheme and worked solutions for 30 questions: 10 multiple choice, 20 structured.

1 · Part 1

Paper 1 & 2 Units [1]

Answer: B

Worked solution

Halve the activity until you reach 800 Bq: 6400 → 3200 (1 half-life) → 1600 (2) → 800 (3). Three half-lives take 90 minutes, so one half-life = 30 minutes. Convert to seconds: 30 min × 60 s/min = 1800 s.

2 · Part 1

Paper 1 & 2 Radioactivity [1]

Answer: D

Explanation

The top number (56) is the mass (nucleon) number and the bottom number (26) is the atomic (proton) number, so the number of neutrons = 56 − 26 = 30. A is the number of protons. B is the total number of nucleons. C adds the two numbers instead of subtracting.

3 · Part 1

Paper 1 & 2 Radioactivity [1]

Answer: A

Explanation

Photographic film is a detector of ionising radiation: it darkens when radiation reaches it, and the amount of darkening shows the dose received. B is wrong because a thin film does not protect the worker. C is wrong because radiation cannot be "cancelled". D is wrong because the film detects all ionising radiation and does not block it.

4 · Part 1

Paper 1 & 2 Radioactivity [1]

Answer: C

Explanation

The patient has radioactive material inside the body, so the patient is contaminated. The food is only exposed to radiation from a source outside it and no radioactive material gets into it, so the food is irradiated (it does not become radioactive). A, B and D mix up the two terms.

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5 · Part 1 Paper 1 & 2 Radioactivity [1]

Answer: A

Explanation

Beta (β−) emission: atomic number +1, mass number unchanged. Alpha emission: atomic number −2, mass number −4. Overall: atomic number +1 − 2 = −1 and mass number −4. B treats beta emission as decreasing the atomic number by 1. C wrongly takes the beta particle as reducing the mass number by 1. D adds the size of the changes in atomic number without the signs.

6 · Part 1 Paper 1 & 2 Radioactivity [1]

Answer: D

Worked solution

Number of half-lives = time ÷ half-life. X: 3 half-lives, so $1600 \div 2^3 = 1600 \div 8 = 200$ Bq. Y: 1 half-life, so $400 \div 2 = 200$ Bq. A sample with a shorter half-life loses activity faster, so the activity of X falls to the same value as the activity of Y.

7 · Part 1 Paper 1 & 2 Fission and fusion [1]

Answer: B

Explanation

Fission, fusion and radioactive decay are all nuclear reactions that release energy and so can be a source of energy (for example, radioactive decay heats the inside of the Earth and powers some space probes). A, C and D each leave out at least one of these.

8 · Part 1 Paper 1 & 2 Fission and fusion [1]

Answer: C

Explanation

The thick concrete is shielding: it absorbs the ionising radiation (such as gamma rays and neutrons) coming from the core so that workers and people nearby are protected. A is wrong because a fission reactor does not use fusion. B is wrong because keeping heat in is not what keeps fission going (neutrons do). D is wrong because nothing can change the rate of radioactive decay.

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9 · Part 1

Paper 1 & 2 | Fission and fusion | [1]

Answer: B

Explanation

When a U-235 nucleus splits, the energy released appears as kinetic energy of the fission products (the daughter nuclei and neutrons fly apart at high speed). This kinetic energy is later transferred to thermal energy in the reactor. A is wrong because fission is a nuclear, not a chemical, process. C and D are not the forms in which the energy is released.

10 · Part 1

Paper 1 & 2 | Fission and fusion | [1]

Answer: A

Explanation

Stars are powered by nuclear fusion, in which small nuclei join to form larger nuclei and energy is released. B is wrong because fission splits heavy nuclei and is not the main process in stars. C is wrong because a star is not burning chemically. D is wrong because radioactive decay is not the main energy source of stars.

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11 · Part 1 Paper 1 & 2 Radioactivity [6]

Mark scheme

(a) **[2]**

- (atoms or nuclei of the same element with) the same number of protons OR the same atomic (proton) number (1)
- different numbers of neutrons OR different mass (nucleon) numbers (1)

(b) **[2]**

- protons: 19 and 19 (1)
- neutrons: 20 for potassium-39 and 21 for potassium-40 (1)

(c) **[1]**

- 19 (1)

(d) **[1]**

- they have different atomic (proton) numbers OR potassium has 19 protons but calcium has 20 protons (1)

> **IGNORE** references to the same mass number

Worked solution

In the symbol ^A_Z X, the top number A is the mass (nucleon) number, the total number of protons and neutrons, and the bottom number Z is the atomic (proton) number. Isotopes are atoms of the same element, so they have the same number of protons, but they have different numbers of neutrons.

(b) Both isotopes have Z = 19, so each nucleus has 19 protons. Neutrons = A − Z: 39 − 19 = 20 and 40 − 19 = 21.

(c) A neutral atom has the same number of electrons as protons, so 19 electrons.

(d) Isotopes must be the same element, which means the same proton number. Potassium-40 has 19 protons and calcium-40 has 20, so they are different elements, even though both have a mass number of 40.

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12 · Part 1 Paper 1 & 2 Radioactivity [6]

Mark scheme

(a) **[2]**

- two protons and two neutrons OR a helium nucleus (1)
- positively charged (charge +2) (1)

(b) **[1]**

- electromagnetic radiation OR an electromagnetic wave (1)

ALLOW has no charge and no mass, if electromagnetic is not stated

(c) **[3]**

- alpha particles are strongly ionising OR gamma rays are weakly ionising (1)
- alpha particles cause many ionisations in a short distance, so they lose their energy quickly (1)
- gamma rays cause few ionisations, so they lose energy slowly and travel much further (1)

Worked solution

(a) An alpha particle is the same as a helium nucleus: 2 protons and 2 neutrons, with a positive charge of +2.

(b) A gamma ray is high-frequency electromagnetic radiation. It has no mass and no charge.

(c) Each time radiation ionises an atom, it loses some energy. Alpha particles are large and highly charged, so they ionise a great many air molecules along their path. They therefore lose all their energy within a few centimetres. Gamma rays ionise very weakly, so they interact with few molecules, lose energy slowly and can travel a long way.

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13 · Part 1 Paper 1 & 2 Radioactivity [7]

Mark scheme

(a) **[1]**

- the type (material) of absorber (1)

ALLOW the absorber OR the material and thickness of the absorber

(b) **[1]**

- 150 − 30 = 120 counts per minute (1)

(c) **[4]**

- alpha is emitted (1)
- because the paper causes a large fall in count rate (960 to 410) (1)
- beta is not emitted, because the aluminium causes almost no further fall (410 to 405, which is within random variation) (1)
- gamma is emitted, because the count rate with lead (150) is still well above background (30) OR the count rate behind aluminium is still well above background (1)

(d) **[1]**

- any one from: handle the source with tongs; keep the source at arm's length OR as far from the body as possible; keep the time of exposure short; return the source to its lead-lined box when not in use; never point the source at anyone (1)

Worked solution

(a) The student changes the absorber, so the absorber (its type) is the independent variable. The count rate is the dependent variable.

(b) The detector also picks up background radiation, so subtract it: 150 − 30 = 120 counts per minute.

(c) Paper stops alpha particles. The count rate falls sharply from 960 to 410 when paper is added, so alpha particles were reaching the detector. A few millimetres of aluminium stops beta particles. The aluminium makes almost no difference (410 to 405; such a small change is just random variation), so there is no beta radiation. Only gamma rays get through paper and aluminium, and they are only partly absorbed by lead. The count rate is still far above the background of 30 counts per minute even with the lead in place, so gamma rays are emitted.

(d) Increasing distance, reducing time and using shielding all reduce the dose received, so handle the source with tongs at arm's length and put it away after use.

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14 · Part 1 Paper 1 & 2 Radioactivity [7]

Mark scheme

(a) **[2]**

- A = 222 (1)
- Z = 86 (1)

(b) **[3]**

Correct equation: $^{214}_{83}Bi \rightarrow {}^{214}_{84}Po + {}^{0}_{-1}e$

- beta particle shown with mass number 0 and atomic number −1 (1)

ALLOW $^{0}_{-1}\beta$

- polonium mass number 214 (1)
- polonium atomic number 84 (1)

(c) **[2]**

- neither the atomic number nor the mass number changes (1)
- because a gamma ray has no charge and no mass OR is electromagnetic radiation (1)

Worked solution

(a) In a nuclear equation, mass numbers and atomic numbers must balance on both sides. Mass: 226 = A + 4, so A = 222. Charge: 88 = Z + 2, so Z = 86.

(b) A beta particle is a fast electron, written $^{0}_{-1}e$. Mass numbers: 214 = A + 0, so A = 214. Atomic numbers: 83 = Z + (−1), so Z = 84. The equation is $^{214}_{83}Bi \rightarrow {}^{214}_{84}Po + {}^{0}_{-1}e$. (In beta decay a neutron changes into a proton, so the atomic number goes up by 1 and the mass number stays the same.)

(c) A gamma ray is a burst of electromagnetic radiation that carries away energy only. It has no mass and no charge, so the nucleus keeps the same atomic number and mass number.

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15 · Part 1 Paper 1 & 2 Radioactivity [6]

Mark scheme

(a) **[2]**

- atomic number does not change (1)
- mass number decreases by 1 (1)

(b) **[4]**

- decrease in mass number = 232 − 220 = 12 (1)
- only alpha emission changes the mass number (by −4), so 12 ÷ 4 = 3 alpha particles (1)
- 3 alpha particles reduce the atomic number by 6, from 90 to 84 (1)
- the final atomic number is 86, so it must rise by 2: 2 beta particles (1)

ALLOW final answer 3 alpha and 2 beta with partial working for the marks shown

Worked solution

(a) A neutron has no charge, so the number of protons, and therefore the atomic number, stays the same. The nucleus loses one nucleon, so the mass number falls by 1.

(b) Use the mass number first, because beta emission does not change it. The mass number falls by 232 − 220 = 12. Each alpha particle removes 4, so there are 12 ÷ 4 = 3 alpha decays. Three alpha decays lower the atomic number by 3 × 2 = 6, giving 90 − 6 = 84. Radon has atomic number 86, which is 2 higher. Each beta (β−) emission raises the atomic number by 1, so there must be 2 beta decays. Check: atomic number 90 − 6 + 2 = 86; mass number 232 − 12 + 0 = 220.

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16 · Part 1 Paper 1 & 2 Radioactivity [6]

Mark scheme

(a) **[2]**

- any two from: radon gas (released from rocks and soil); rocks such as granite OR the ground; building materials; food and drink OR radioactive isotopes in our bodies; medical uses such as X-rays; nuclear power stations OR nuclear waste OR fallout from nuclear weapons tests (1) (1)

(b) **[1]**

- cosmic rays OR radiation from the Sun OR radiation from stars (1)

(c) **[2]**

- (18 + 23 + 20 + 25 + 19) ÷ 5 = 105 ÷ 5 (1)
- 21 counts per minute (1)

ALLOW counts/min OR cpm

(d) **[1]**

- radioactive decay is random (1)

Worked solution

(a) Most background radiation comes from natural sources on Earth, for example radon gas that seeps out of rocks such as granite, and radioactive isotopes in rocks, soil, building materials and food. Some comes from human activity, such as medical X-rays and nuclear waste.

(b) Cosmic rays reach the Earth from the Sun and from outer space.

(c) Mean = total ÷ number of readings = 105 ÷ 5 = 21 counts per minute.

(d) It is impossible to predict when a particular nucleus will decay. The number of decays in each minute therefore varies randomly about an average value, even when nothing changes.

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17 · Part 1 Paper 1 & 2 Radioactivity [7]

Mark scheme

(a) **[2]**

- the time taken for half (1)
- of the undecayed nuclei to decay OR the activity to fall to half its initial value (1)

(b) **[2]**

- activity halves from 720 Bq to 360 Bq in 4 minutes OR 360 Bq to 180 Bq from 4 to 8 minutes (1)
- half-life = 4 minutes (1)

ALLOW 4 min OR 240 s

(c) **[2]**

- 16 minutes = 4 half-lives OR 720 → 360 → 180 → 90 → 45 (1)
- 45 Bq (1)

ALLOW ECF from the half-life in (b)

(d) **[1]**

- the number of undecayed (radioactive) nuclei decreases OR there are fewer nuclei left to decay (1)

Worked solution

(a) The half-life is the time taken for half of the undecayed nuclei in a sample to decay, which is also the time for the activity to fall to half its value.

(b) From the table, the activity falls from 720 Bq to 360 Bq (half) between 0 and 4 minutes, and from 360 Bq to 180 Bq between 4 and 8 minutes. Both give a half-life of 4 minutes.

(c) 16 ÷ 4 = 4 half-lives. Halve four times: 720 → 360 → 180 → 90 → 45 Bq.

(d) Each decay uses up an unstable nucleus. As time passes there are fewer undecayed nuclei, so fewer decays happen each second and the activity falls.

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18 · Part 1 Paper 1 & 2 Radioactivity [6]

Mark scheme

(a) **[1]**

- 1630 − 30 = 1600 counts per minute (1)

(b) **[3]**

- 12 ÷ 3.0 = 4 half-lives (1)
- count rate from source = 1600 ÷ 2^4 = 1600 ÷ 16 = 100 counts per minute (1)
- detector reading = 100 + 30 = 130 counts per minute (1)

ALLOW ECF from (a)

(c) **[2]**

- count rate from source = 55 − 30 = 25 counts per minute, and 1600 → 800 → 400 → 200 → 100 → 50 → 25 is 6 half-lives (1)
- 6 × 3.0 = 18 hours (1)

ALLOW ECF from (a)

Worked solution

The detector always measures background radiation as well as radiation from the source. Only the part due to the source halves every half-life, so remove the background first and add it back at the end.

(a) Source alone: 1630 − 30 = 1600 counts per minute.

(b) 12 hours ÷ 3.0 hours = 4 half-lives. The source's count rate becomes 1600 ÷ 16 = 100 counts per minute. The detector also picks up the 30 counts per minute of background, so it reads 130 counts per minute. (A common mistake is to halve 1630 four times, giving about 102.)

(c) A reading of 55 counts per minute means the source alone gives 55 − 30 = 25 counts per minute. 1600 ÷ 25 = 64 = 2^6, so 6 half-lives have passed: 6 × 3.0 = 18 hours.

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19 · Part 1 Paper 1 & 2 Radioactivity [5]

Mark scheme

(a) **[2]**

- 24 ÷ 8 = 3 half-lives OR $6.4 \times 10^{14} \rightarrow 3.2 \times 10^{14} \rightarrow 1.6 \times 10^{14} \rightarrow 8.0 \times 10^{13}$ (1)
- 8.0×10^{13} (1)

(b) **[1]**

- $6.4 \times 10^{14} - 8.0 \times 10^{13} = 5.6 \times 10^{14}$ (1)

> **ALLOW** ECF from (a)

(c) **[2]**

- in each half-life only half of the remaining nuclei decay (1)
- after 16 days (2 half-lives) one quarter remain OR 1.6×10^{14} nuclei are still undecayed (1)

Worked solution

(a) 24 days ÷ 8 days = 3 half-lives. Halve three times: $6.4 \times 10^{14} \rightarrow 3.2 \times 10^{14} \rightarrow 1.6 \times 10^{14} \rightarrow 8.0 \times 10^{13}$ undecayed nuclei. (That is 1/8 of the original number.)

(b) Number decayed = start − left = $6.4 \times 10^{14} - 0.8 \times 10^{14} = 5.6 \times 10^{14}$.

(c) The student thinks two "halves" make a whole. In fact, each half-life halves the number of nuclei that are left, not the original number. After 8 days half remain; after 16 days half of that half, one quarter, remain: 1.6×10^{14} nuclei. The number keeps halving and never quite reaches zero in a fixed number of half-lives.

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20 · Part 1 Paper 1 & 2 Radioactivity [7]

Mark scheme

(a) **[3]**

- gamma rays are very penetrating, so they pass out of the body and reach the detector (1)
- alpha particles would be absorbed inside the body and would not reach the detector (1)
- alpha particles are strongly ionising, so they would cause more damage to cells inside the body OR gamma rays are weakly ionising, so they cause less damage (1)

(b) **[2]**

- long enough for the tracer to spread through the body and for the scan to be done before the activity becomes too low (1)
- short enough that the activity falls quickly afterwards, so the patient receives a small dose of radiation (1)

(c) **[2]**

- radiotherapy (1) – gamma rays OR radiation from a source are used to kill cancer cells (1)
- OR sterilising medical equipment (1) – gamma rays kill bacteria on the sealed equipment (1)
- OR any other valid medical use (1) with a correct description of how the radiation is used (1)

Worked solution

(a) The detector is outside the body, so the radiation must get out of the body to be detected. Gamma rays pass easily through tissue. Alpha particles would be stopped within a fraction of a millimetre of tissue and never reach the detector. Alpha particles are also strongly ionising, so all their energy would be dumped in nearby cells, causing a lot of damage.

(b) A tracer must stay active long enough for doctors to take the images (a few hours). It should then decay quickly so the patient is not exposed to radiation for longer than necessary. A half-life of seconds would be too short to use; a half-life of years would expose the patient for a very long time.

(c) For example, in radiotherapy a beam of gamma rays is aimed at a tumour to kill the cancer cells. Medical instruments can also be sterilised by gamma rays, which kill bacteria even through the sealed packaging.

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21 · Part 1 Paper 1 & 2 Radioactivity [7]

Mark scheme

(a) **[3]**

- alpha particles would be completely absorbed by the paper, so no count (or no change in count) would be detected (1)
- gamma rays would pass through the paper almost unaffected, so the count rate would hardly change when the thickness changes (1)
- beta particles are partly absorbed by paper, so the count rate depends on (changes with) the thickness of the paper (1)

(b) **[2]**

- the paper has become thinner (1)
- the gap between the rollers should be made larger (1)

> **ALLOW** the rollers should be moved apart

(c) **[2]**

- the activity of the source would fall (noticeably), so the count rate would fall even though the thickness has not changed (1)
- the computer would treat the lower count rate as paper that is too thick, and would make the gap smaller, so the paper becomes thinner (1)

Worked solution

(a) The system only works if the count rate changes when the thickness changes. Alpha particles cannot get through even a sheet of paper, so the detector would read nothing whatever the thickness. Gamma rays pass straight through paper, so a small change in thickness would make almost no difference. Beta particles are partly stopped by paper: thicker paper absorbs more of them, thinner paper absorbs fewer. So the count rate is a sensitive measure of thickness.

(b) A higher count rate means less radiation is being absorbed, so the paper is thinner than it should be. To make thicker paper, the computer must open the gap between the rollers.

(c) With a short half-life, the activity of the source falls noticeably over days or weeks. The count rate falls for this reason alone. The computer cannot tell the difference, so it "thinks" the paper is too thick and closes the gap, making the paper too thin. A source with a long half-life gives a steady count rate for years.

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22 · Part 1 Paper 1 & 2 Radioactivity [7]

Mark scheme

(a) **[2]**

- contamination: radioactive material gets onto or into an object or person (1)
- irradiation: an object or person is exposed to radiation from a source outside it (and does not become radioactive) (1)

(b) **[3]**

- outside the body, alpha particles are stopped by the air or by the outer layer of skin, so they do not reach living cells (1)
- inside the body, the source is right next to living cells (tissue) (1)
- alpha particles are strongly ionising, so they cause a lot of damage to the cells near the source (1)

(c) **[2]**

- radiation can damage OR kill cells and tissue (1)
- radiation can cause mutations (in DNA), which may lead to cancer (1)

Worked solution

(a) Contamination means that radioactive atoms have got onto or into something, for example by touching, swallowing or breathing in radioactive material; the contaminated object then keeps emitting radiation. Irradiation means that something has been exposed to radiation from a source outside it; it does not become radioactive itself.

(b) Alpha particles have a very short range: a few centimetres of air or the dead outer layer of skin stops them, so a source outside the body does little harm. If the source is swallowed or breathed in, it sits right next to living cells. Alpha particles are the most strongly ionising radiation, so all their energy is deposited in the nearby cells, causing serious damage.

(c) Ionising radiation can kill or damage cells and tissue (as in radiation burns), and it can change the DNA in cells, causing mutations that may lead to cancer.

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23 · Part 1 Paper 1 & 2 Radioactivity [7]

Mark scheme

(a) **[2]**

- $1/16 = (1/2)^4$, so 4 half-lives OR $1 \rightarrow 1/2 \rightarrow 1/4 \rightarrow 1/8 \rightarrow 1/16$ (1)
- $4 \times 30 = 120$ years (1)

(b) **[5]**

Indicative points, any five (1) each. For full marks, the answer must include at least one problem and at least one way of reducing the risk.

Problems:

- some waste stays radioactive for a very long time (thousands of years) because it contains isotopes with long half-lives (1)
- radioactive material could leak out and get into groundwater, water supplies or the food chain (1)
- people could be exposed to ionising radiation, which damages cells and causes mutations OR cancer (1)
- waste could be stolen or misused OR there could be accidents while it is transported (1)
- the storage site must be monitored for many future generations (1)

Reducing the risk:

- seal the waste in strong containers, for example steel or concrete, OR turn it into glass (1)
- store it deep underground in stable rock (1)
- choose sites away from people and from water supplies OR in areas without earthquakes (1)
- monitor the storage site and its surroundings regularly (1)
- use shielding (for example concrete or lead) and handle waste remotely (1)

Worked solution

(a) Each half-life halves the activity: $1 \rightarrow 1/2 \rightarrow 1/4 \rightarrow 1/8 \rightarrow 1/16$ takes 4 half-lives. Time = 4×30 years = 120 years.

(b) The main problem is time. Waste containing isotopes with long half-lives stays dangerous for thousands of years, far longer than any building lasts. If containers leak, radioactive material can get into groundwater and food, and people who take it in are contaminated; the radiation damages cells and can cause cancer. Waste must also be kept safe from theft and from accidents during transport, and someone must monitor it for many generations. The risks are reduced by sealing the waste in steel or concrete containers (or melting it into glass), burying it deep underground in stable rock far from people and water supplies, shielding and handling it remotely, and monitoring the site.

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24 · Part 1 Paper 1 & 2 Fission and fusion [5]

Mark scheme

(a) **[2]**

- mass numbers: 1 + 235 = 141 + 92 + x OR 236 = 233 + x (1)
- x = 3 (1)

(b) **[3]**

- each fission releases (two or three) neutrons (1)
- these neutrons strike (are absorbed by) other U-235 nuclei (1)
- causing those nuclei to undergo fission, which release more neutrons, and so on (1)

Worked solution

(a) The total mass number must be the same on both sides. Left: 1 + 235 = 236. Right: 141 + 92 + x × 1 = 233 + x. So x = 236 − 233 = 3. Check the atomic numbers: left 0 + 92 = 92, right 56 + 36 + 0 = 92, which balances.

(b) Each fission releases three neutrons in this case. If these neutrons hit other U-235 nuclei, those nuclei split too, each releasing more neutrons, which can split still more nuclei. Each fission can therefore trigger further fissions: this is a chain reaction. In a large piece of uranium few neutrons escape, so the chain can keep going.

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25 · Part 1 Paper 1 & 2 Fission and fusion [7]

Mark scheme

(a) **[2]**

- the moderator slows down the neutrons (released by fission) (1)
- slow neutrons are more likely to be absorbed by U-235 nuclei and cause fission (1)

(b) **[2]**

- control rods absorb neutrons (1)
- lowering the rods further into the core absorbs more neutrons, so fewer fissions happen; raising them absorbs fewer, so more fissions happen (1)

ALLOW the rods control the number of neutrons available to cause fission

(c) **[2]**

- factor = 1100 ÷ 1000 = 1.1 (1)
- generation 7: 1331 × 1.1 = 1464 (1)

ALLOW 1460 OR 1464.1

(d) **[1]**

- lower the control rods further into the core (1)

Worked solution

(a) Neutrons released by fission are moving very fast. The moderator slows them down, and slow neutrons are far more likely to be absorbed by U-235 nuclei and make them split, so the chain reaction can continue.

(b) Control rods absorb neutrons. Pushing them deeper into the core soaks up more neutrons, leaving fewer to cause fission, so the rate of fission falls. Pulling them out lets more neutrons cause fission, so the rate rises.

(c) Before the change the number of fissions is steady at 1000 per generation. Afterwards: 1100 ÷ 1000 = 1.1, 1210 ÷ 1100 = 1.1 and 1331 ÷ 1210 = 1.1, so the number grows by a factor of 1.1 each generation. Generation 7: 1331 × 1.1 = 1464 fissions.

(d) The rate is increasing, so too few neutrons are being absorbed (the rods must have been raised). Lowering the rods back into the core absorbs more neutrons until the number of fissions per generation stops rising.

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26 · Part 1 Paper 1 & 2 Fission and fusion [5]

Mark scheme

(a) **[3]**

- mass number = 2 + 3 − 4 = 1 (1)
- atomic number = 1 + 1 − 2 = 0 (1)
- X is a neutron (1)

ALLOW $^{1}_{0}$n for the third mark

(b) **[2]**

- the lost mass has been converted into (released as) energy (1)
- so the reaction releases energy OR is a source of energy (1)

REJECT a bare restatement that mass has been lost, with no link to energy

Worked solution

(a) Balance the mass numbers: 2 + 3 = 5 on the left, and 4 + A on the right, so A = 1. Balance the atomic numbers: 1 + 1 = 2 on the left, and 2 + Z on the right, so Z = 0. A particle with mass number 1 and no charge is a neutron, $^{1}_{0}$n.

(b) In fusion, the larger nucleus formed has less mass than the small nuclei that joined to make it. This loss of mass is accompanied by the release of energy, which is why fusion can be a source of energy.

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27 · Part 1 Paper 1 & 2 Fission and fusion [5]

Mark scheme

(a) **[4]**

- nuclei are positively charged OR contain protons (1)
- (so) nuclei repel each other (electrostatic repulsion) (1)
- at low temperatures nuclei move too slowly OR do not have enough kinetic energy to overcome the repulsion (1)
- so they cannot get close enough to fuse (1)
- at low pressure the nuclei are far apart, so collisions are too rare (1)
- max 4

(b) **[1]**

- any one from: it is very difficult to contain material at such a high temperature (no container can withstand it); a large amount of energy is needed to heat the hydrogen to this temperature; it is difficult to keep the high temperature and pressure for long enough (1)

Worked solution

(a) Fusion needs two nuclei to get extremely close. All nuclei contain protons, so they are positively charged and repel each other. At ordinary temperatures the nuclei move too slowly and do not have enough kinetic energy to overcome this electrostatic repulsion, so they never get close enough to fuse. Very high temperatures make the nuclei move fast enough to overcome the repulsion, and high pressure packs them close together so that collisions happen often.

(b) Material at more than 100 million °C would melt or damage any ordinary container, so it has to be held away from the walls; heating it and keeping it this hot also takes a lot of energy.

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28 · Part 1 Paper 1 & 2 Fission and fusion [5]

Mark scheme

(a) **[4]**

Indicative points, any four (1) each. For full marks, both processes must be described.

- fission: a large (heavy) nucleus such as U-235 (1)
- splits into two smaller (daughter) nuclei (and some neutrons) (1)
- fusion: two small (light) nuclei such as hydrogen (1)
- join to form a larger nucleus (1)
- fission is started when a nucleus absorbs a neutron, but fusion needs very high temperatures and pressures (1)
- the products of fission are radioactive, but the main product of fusion (helium) is not (1)

(b) **[1]**

- both release energy OR both are nuclear reactions (involving changes to nuclei) (1)

Worked solution

(a) In fission, a large nucleus such as uranium-235 absorbs a neutron and splits into two smaller daughter nuclei, also releasing a few neutrons. In fusion, two small nuclei, such as isotopes of hydrogen, join together to make one larger nucleus, such as helium. Fission can happen at ordinary temperatures once a neutron is absorbed; fusion only happens at very high temperatures and pressures. Fission products are radioactive, while helium from fusion is not.

(b) Both are nuclear reactions in which nuclei change, and both release energy.

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29 · Part 1 Paper 1 & 2 Fission and fusion [6]

Mark scheme

(a) **[2]**

- table-tennis balls represent neutrons (1)
- mousetraps represent uranium-235 nuclei (1)

ALLOW nuclei OR atoms of uranium (fuel) for the mousetraps

(b) **[2]**

- doubling at each stage: 1, 2, 4, 8, 16 OR 2^4 (1)
- 16 (1)

(c) **[2]**

- control rods (1)
- because they absorb neutrons (balls) so that fewer fissions (traps springing) happen (1)

REJECT moderator (it slows neutrons, it does not absorb them) OR shielding (it absorbs radiation leaving the core, not neutrons within the chain reaction)

Worked solution

(a) A springing mousetrap is like a U-235 nucleus undergoing fission, and the balls it throws out are like the neutrons released. When a ball lands on another trap, it is like a neutron causing another fission.

(b) Each trap that springs sets off two more, so the number doubles at each stage: stage 1 = 1, stage 2 = 2, stage 3 = 4, stage 4 = 8, stage 5 = 16.

(c) The card catches balls so that they cannot set off more traps. In a reactor, control rods do the same job: they absorb neutrons so that fewer nuclei undergo fission and the chain reaction is controlled.

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30 · Part 1 Paper 1 & 2 Fission and fusion [6]

Mark scheme

(a) **[2]**

- energy = $2.5 \times 10^{21} \times 3.2 \times 10^{-11}$ (1)
- 8.0×10^{10} J (1)

(b) **[2]**

- mass = $8.0 \times 10^{10} \div (3.0 \times 10^{7})$ (1)
- 2.7×10^{3} kg (1)

ALLOW 2667 kg OR 2670 kg; ECF from (a)

(c) **[2]**

- fission produces daughter nuclei (fission products) (1)
- the daughter nuclei are radioactive OR emit ionising radiation (1)

Worked solution

(a) Total energy = number of fissions × energy per fission = $2.5 \times 10^{21} \times 3.2 \times 10^{-11}$ J = 8.0×10^{10} J.

(b) Each kilogram of coal gives 3.0×10^{7} J, so the mass needed = $8.0 \times 10^{10} \div (3.0 \times 10^{7})$ = 2667 kg, about 2.7×10^{3} kg. A small amount of nuclear fuel releases as much energy as nearly three tonnes of coal.

(c) Each fission produces two daughter nuclei. These are radioactive, so the used fuel emits much more ionising radiation than the fresh fuel and must be handled remotely and shielded.

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